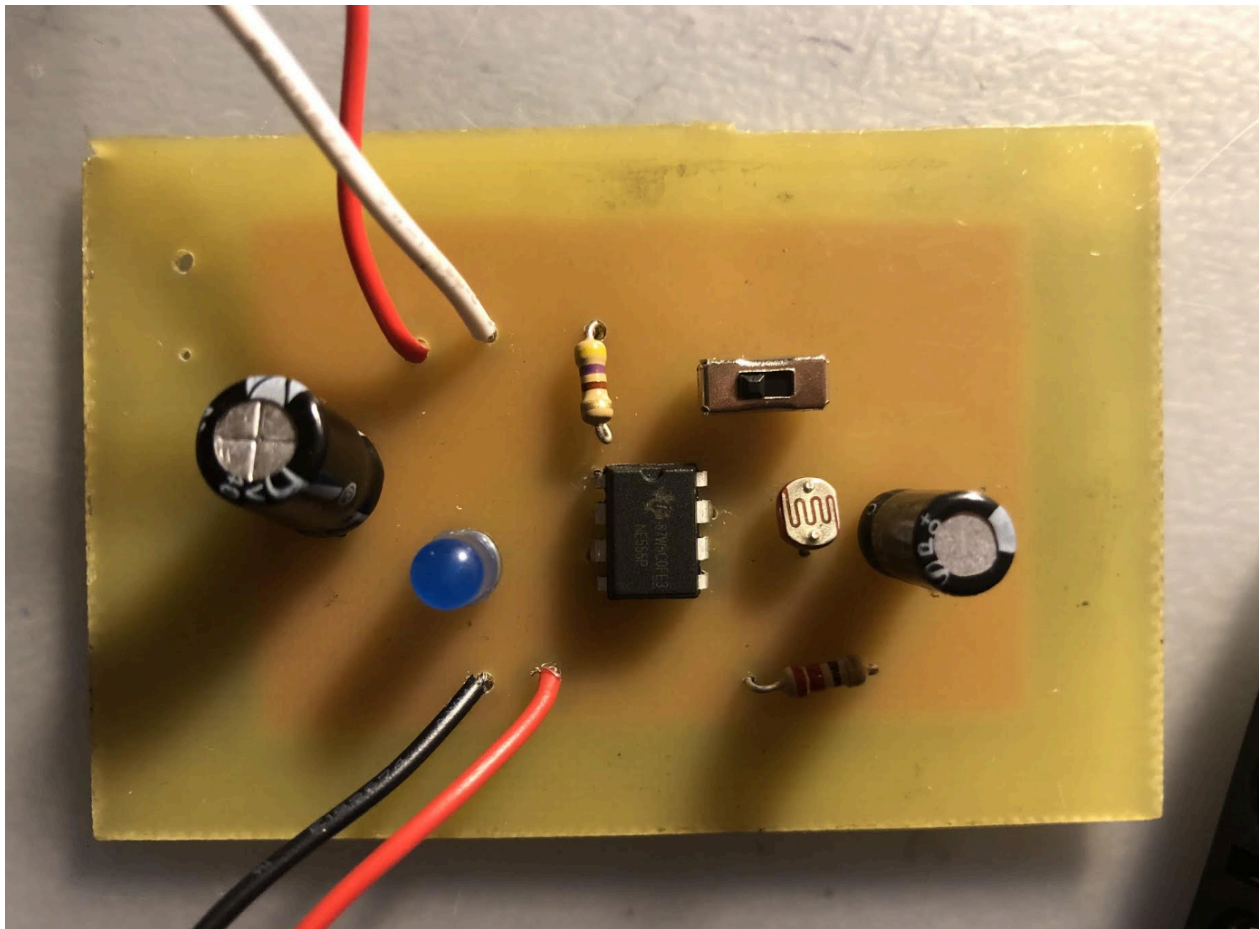


2.2.5 PCB Design

Flashing Light Sensor



Honors Digital Electronics Period 6
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January 9, 2023 - February 16, 2023

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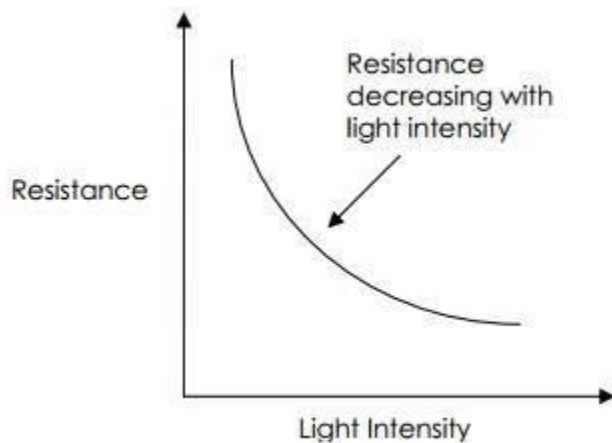
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Design Specifications

Circuit Design:

An LDR, or light dependent resistor, is a component that decreases in resistance with higher light intensity, according to Kitronik.com. Based on specific wavelengths of light, the resistor will react and can be connected to other components to manage the function of a circuit. They are usually applied as light sensors in circuits; when light is below a threshold intensity an LED will remain off, and if it is above, it will switch on. I hoped to use the oscillations of a 555 timer to reflect the intensity of light based on an LDR, so I looked for a circuit schematic that followed this design.

Figure A: Light Intensity vs Resistance from Kitronik.com



Initially, I had a simple circuit with an LED, two resistors, a transistor, and an LDR, but I decided to implement a 555 timer and speaker to give my circuit a timed aspect. This would allow for a more informative output. Based on the brightness of light shined on the LDR, the LED would fluctuate more, indicating the intensity of light. My design was mainly based off of [Electronzap's](#) Youtube video, "Light dependent resistor LDR astable 555 timer with speaker circuit learning electronics lesson 0064" and the schematic provided in the video. He uses an LDR, 555 timer, resistors, 2 capacitors, 1 LED, and a speaker to create an output. The speaker's ticks speed up with more intense light, alongside the flashing LED.

Intended Use

I intend to use this circuit to monitor the light in certain areas of my house and ensure that the lights are not too bright. My PCB can be helpful in power conservation since my house has dimming lights, so if the LED flashes more frequently, the light is most likely too bright and I need to reduce its intensity. Outside of my house, it can be used in a variety of public spaces to indicate how bright a light should be to save electricity.

Schematic and Function

My initial schematic was primarily based on Electronzap's 555 Timer and LDR Circuit. While prototyping the breadboard, however, a major issue I had was that there was no "off" state. This meant the circuit was constantly active, which was quite inconvenient, so I implemented a switch based on my own testing. The schematic below is directly from Electronzap. I attached a switch that was connected to power to pin 8 on the 555 timer and disconnected pin 8's original connection to pin 4. When I tested this on a breadboard, this setup worked and I was able to turn the breadboard on and off. I also omitted the second LED that was connected to power because I did not want to have more than 10 components on my PCB for the sake of ease. My edits are shown below in Figure B.

The LDR receives a specific intensity of light. Based on that amount of light, it allows a certain amount of voltage to reach the capacitor. Its resistance affects how fast the capacitor discharges and charges, and essentially acts as the threshold trigger resistor in a 555 timer circuit. The capacitor discharges through pins 2 and 6, and this provides power to pin 3 so that the LEDs will flash. The flashing is based on the charging and discharging of the capacitor through the astable 555 timer. An astable 555 has no user control and utilizes two external resistors and one capacitor. To power the entire circuit, pin 8 and 4 are connected to a switch, which is connected to power. When the switch is turned on, power is supplied, and the entire circuit can function as intended.

For this circuit, I was unable to test it with a Multisim circuit because the simulation did not have an LDR/photoresistor component. For this reason, I decided to breadboard it directly and test specific components that worked well. Along the way, I changed the resistor values to and capacitor values in the schematic because the physical circuit functioned differently than the schematic .

Figure A:
Original
Schematic

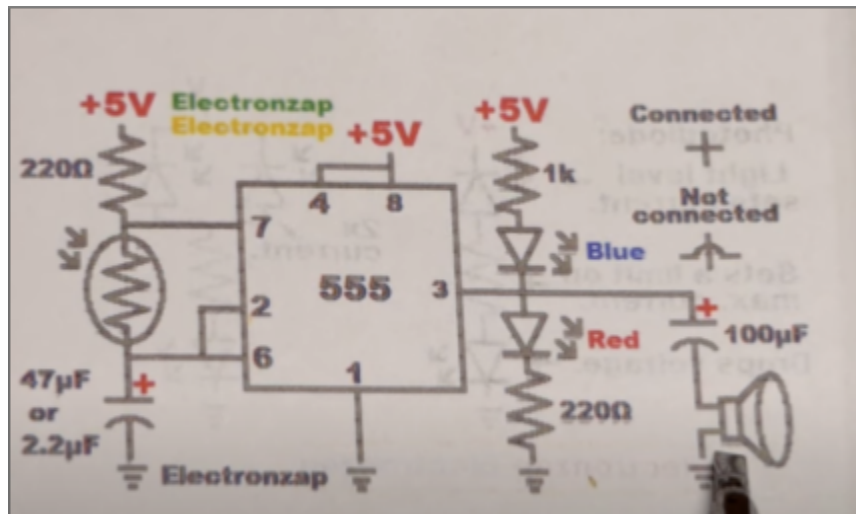
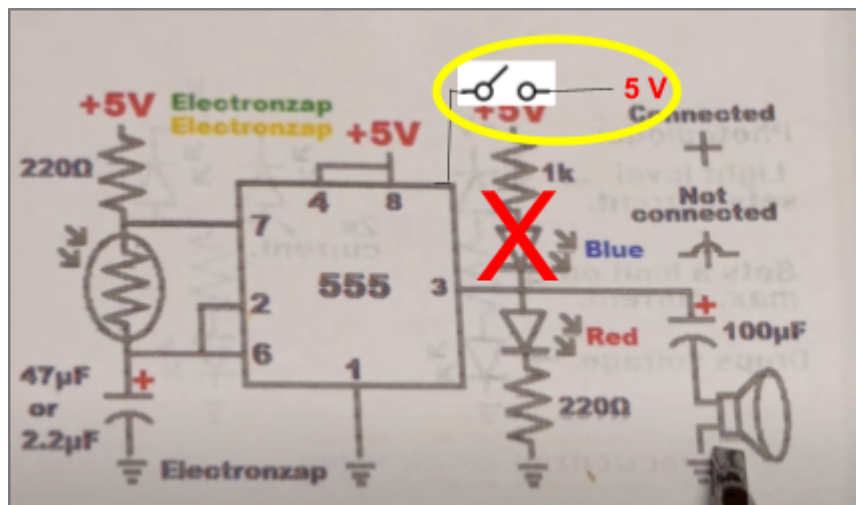
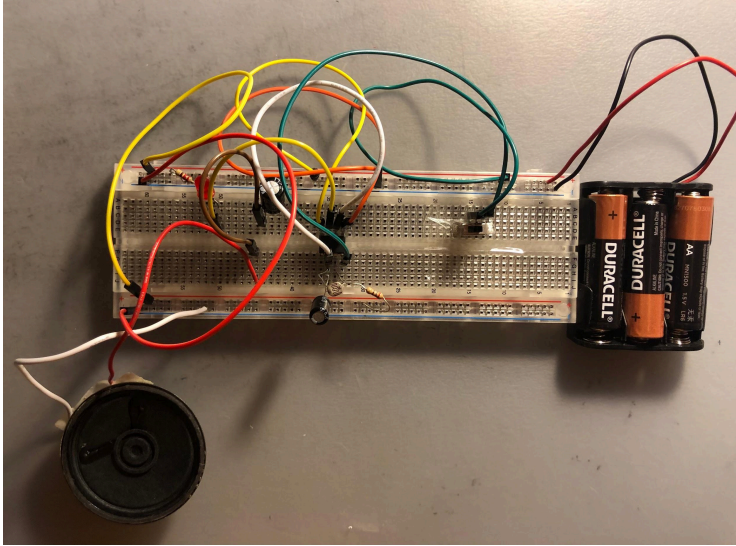


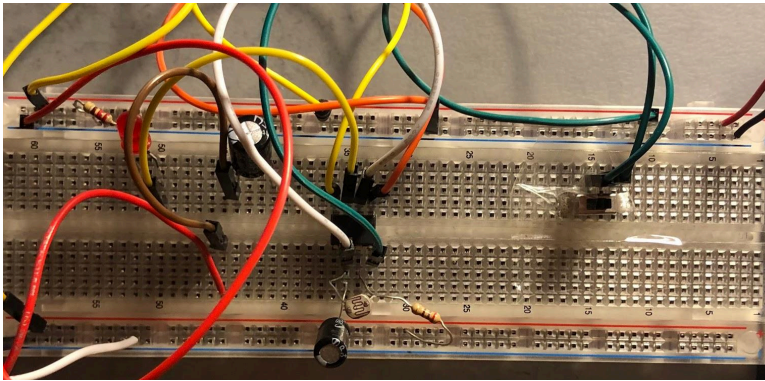
Figure B:
SPDT Switch
Added to Pin 8,
LED omitted



Breadboard Prototype

Below are images from multiple views of the breadboarded circuit I used and why I chose specific components.

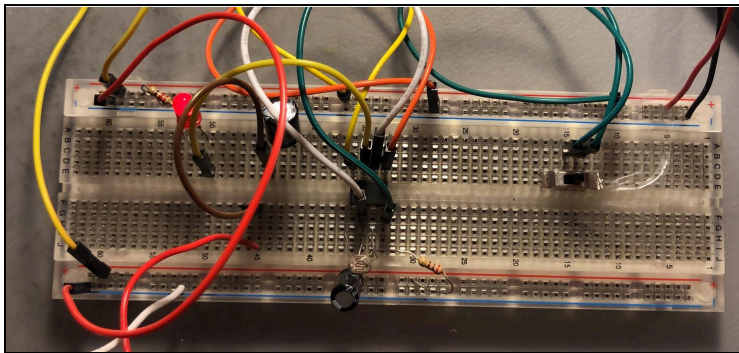
Image	Description
	General top view The circuit has 10 components: ❖ 1 LED ❖ 1 555 Timer ❖ 1 SPDT slide switch ❖ 1 47 uF capacitor ❖ 1 220 uF capacitor ❖ 2 220 ohm resistors ❖ 1 4.5 V battery pack ❖ 1 25 ohm speaker ❖ 1 LDR (light dependent resistor) The circuit overall is quite compact and during testing, it was very easy to test different capacitor values, resistors, and timer setups. Through troubleshooting and testing, I was also able to implement a switch to power the entire circuit in general.



Close up

There is a 555 timer, located in the center. The switch is connected to pin 8, the LDR between pins 6 and 7, and a capacitor through pin 6 as well. Pin 5 can be grounded or left empty since it has no function in this circuit, so I did not add anything.

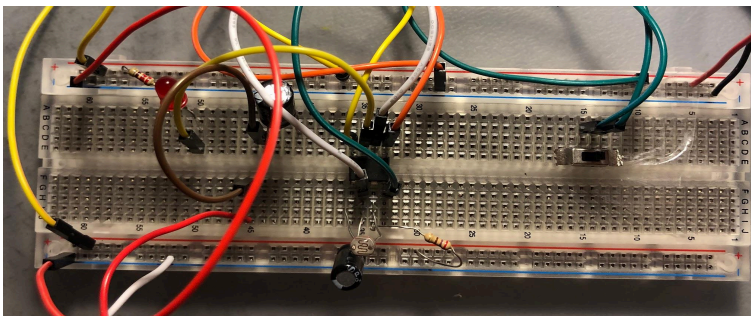
Pin 1 is connected to ground and pin 2 is connected to pin 6. Pin 3 is a very important pin because it activates the LED output and the speaker output. A 220 uF capacitor is connected to pin 3 and this powers a speaker that is grounded. Additionally, pin 3 connects to the anode of an LED, which is then grounded. Pin 4 is grounded.



“On” State

This image captures the “on” phase of the circuit, where the LED is flashing based on the intensity of light.

(Video of circuit functioning in P225 Folder)



“Off” State

This image captures the “off” phase of the circuit, where the LED is not flashing at all.

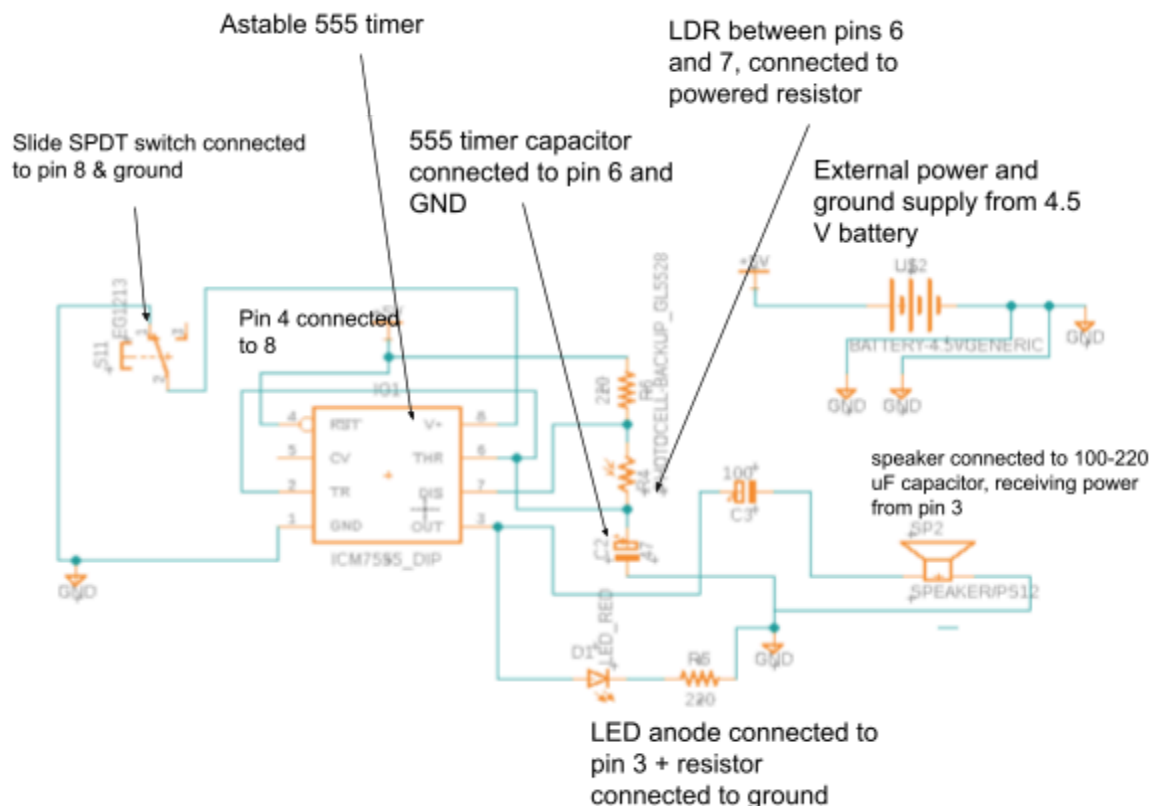
Fabrication Design

To design a physical PCB, I used Fusion 360 to create a schematic, PCB mask, and 3D model.

Schematic

For the schematic, I arranged it similar to the one from Electronzap's video (from the previous section), except I included the edits I had made on the physical breadboard. This included the switch and removal of the second LED. I could not test it so I made sure it matched as closely as possible to the physical prototype I had made. It took time to find many components with the correct footprints because there were hundreds of libraries on Fusion 360, but after comparing measurements and footprints, I was eventually able to create a useful schematic. In my schematic, there is a NE555 timer, Photoresistor (LDR), 5 mm LED, capacitors, and speaker. I also included a battery pack and speaker, but these components would not have large footprints because I planned to simply solder their wires into the holes, as opposed to physically attaching them to the board.

Figure A: Fusion 360 Schematic

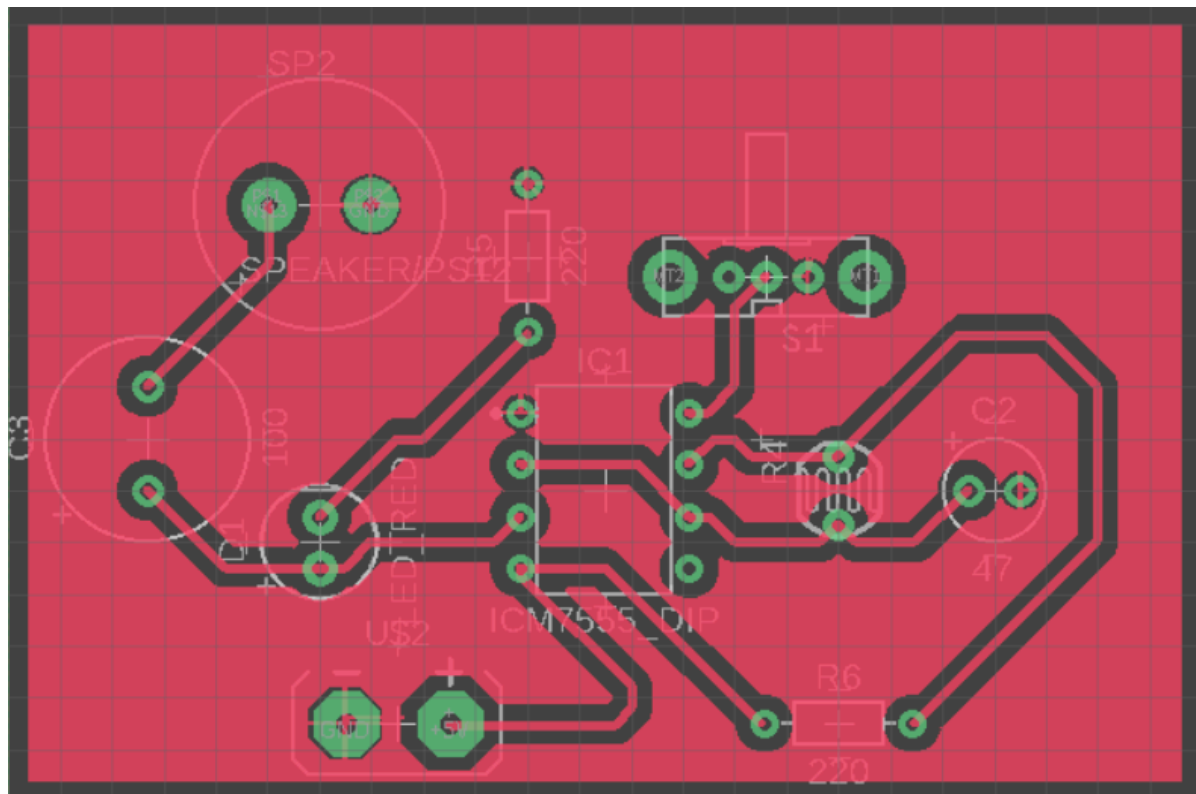


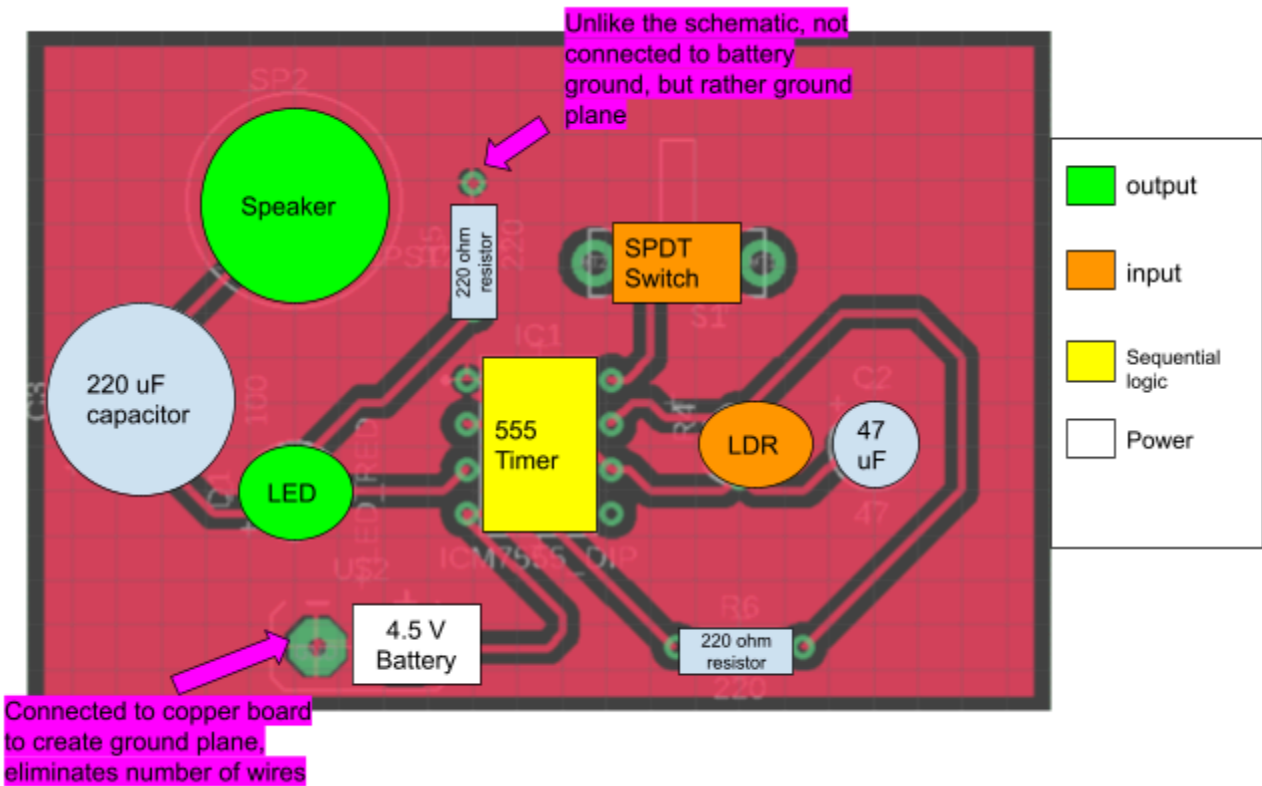
PCB Design

Fusion 360 has a helpful feature that allows the user to create a schematic and link it to a PCB document. This makes it easier to route wires because there are already “airwires” in place that connect the components. The components have the correct footprint, so the PCB mask is realistic and useful in creating the physical board.

After finalizing the schematic above, the PCB was made by linking the two documents. Based on the airwires, I manually routed the components to the optimal layout. I managed to avoid using any jumper/via connections under the board with my layout. I added a ground plane to avoid making several individual connections to ground. A ground plane is a surface connected to ground that provides a strong ground connection for components. My layout was mainly focused on avoiding jumper connections but I tried to have all my inputs on the right (switch and LDR) and outputs on the right (speaker and LED). I increased my isolation to 0.025 mm and had 0.0024 mm thickness wires. Since I had fewer components, my board was 1.5” x 2”.

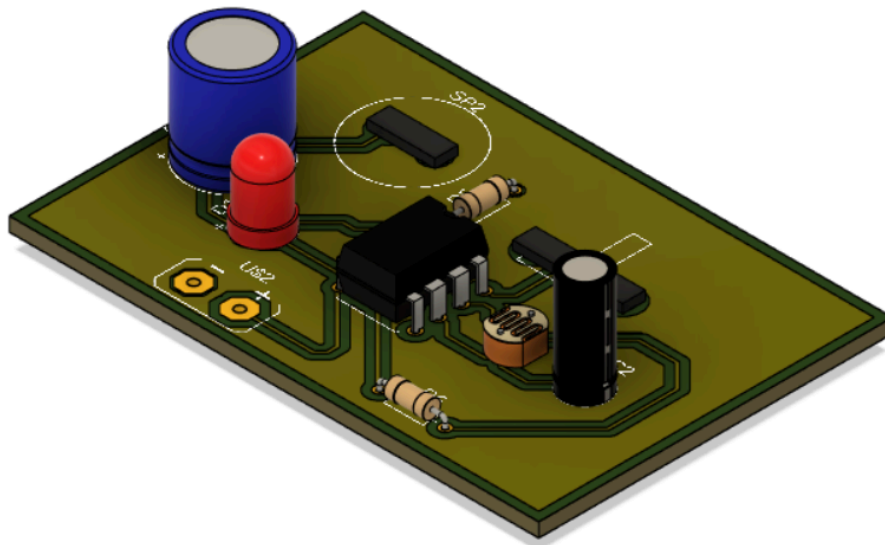
Figure B: Fusion 360 PCB + Annotated Diagram





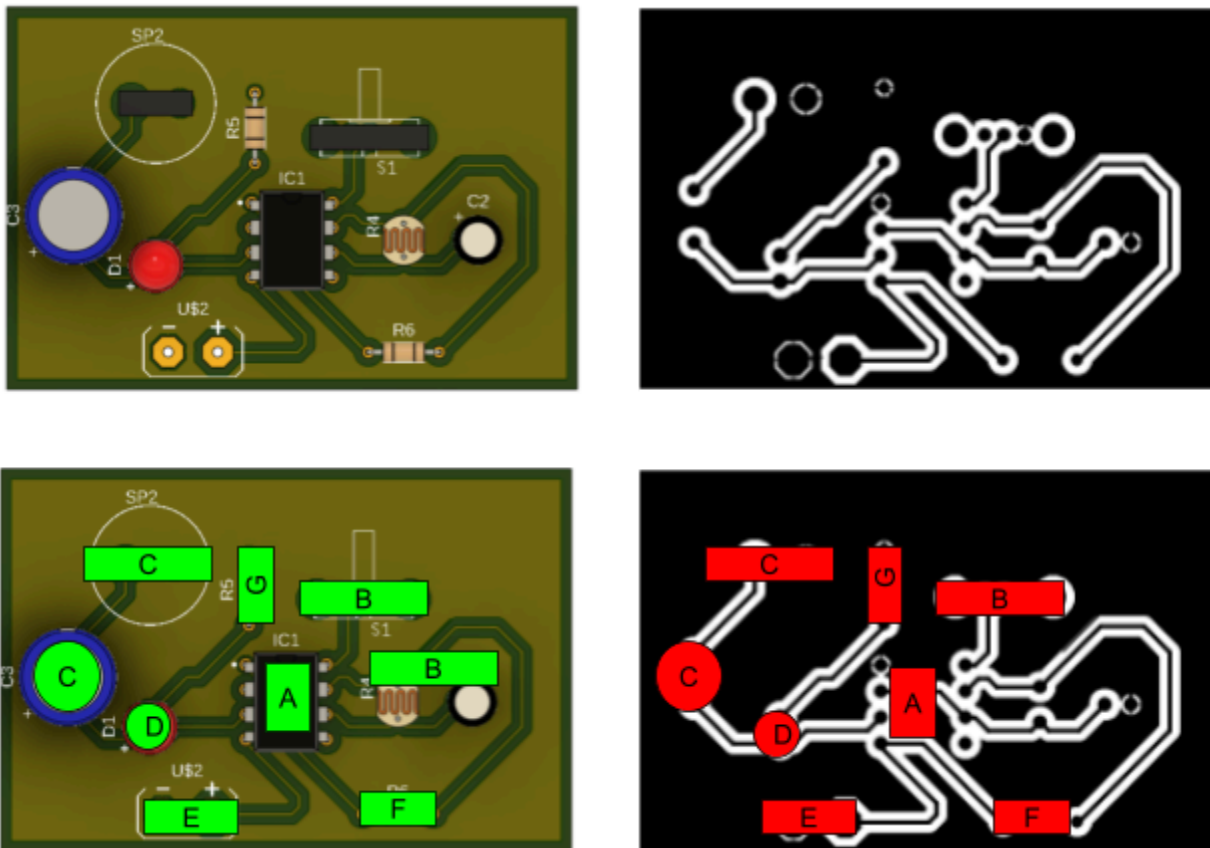
After creating my PCB, I used Fusion 360 to link a 3D model to it. This would give me an idea of the spacing and whether I used the right components.

Figure C: 3D Model



To create my physical board, I needed a black and white mask layer. Using Viewmate, I exported the “Copper Top” Gerber Image of the PCB. Compared to the physical model and the PCB, it looked accurate, so I decided to begin the steps of masking it using the iron and copper board.

Figure D: 3D Model and Mask Side by Side Comparison



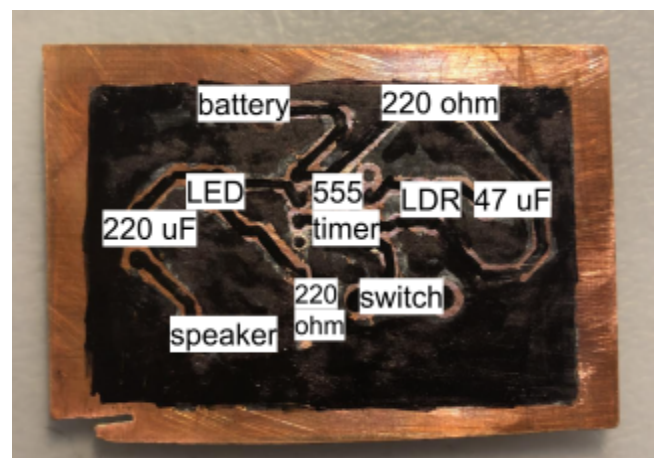
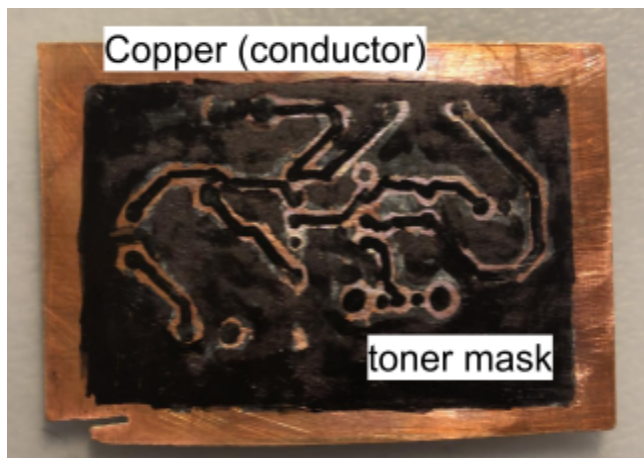
Final Copper Mask

Once I had finalized my glossy paper mask and ensured that it was to scale by testing components in it, I began the tedious process of masking. Masking was a very difficult step for me because I had a ground plane and initially, the paper we had used for masking was too thick. I ironed several iterations for up to 40 minutes, but none were successful, because there was not enough heat and compression to transfer the toner onto the board.

As a class this was an issue, so thin magazine paper was brought in to make the process easier. It did make it easier to a certain extent but I still had to hold down the iron for up to 30 minutes due to heat and compression issues. Moreover, my print was dark, especially the ground plane and the wires, so it took time to transfer properly. Most of the time, I had to redo it because only a small part would be blotchy.

As a backup method for masking, I tried using acetone. I diluted the acetone solution and poured it on top of the board with the magazine paper. Then, I began rubbing and applying equal pressure to all the areas to make sure that I wasn't damaging the paper or ruining the print. The quality of the etching was higher quality for some attempts, however patches of toner would not transfer on properly, so I decided to stick with ironing. In the end, I managed to get two copper boards with masks on them using an iron.

Figure E: Annotated Copper Mask



Etching and Drilling

I used Ferric Chloride to eat away at the copper on the board and leave only the copper under my mask. I dipped it in the solution for 30 minutes and ensured that the correct parts without the toner mask were being eaten away.

Once I had fully etched the board and ensured my wires were isolated, I began the drilling process. I used a 0.7 and 0.8 mm drill bit for most of the holes in my circuit because they were larger. For the resistors and the LEDs, the holes were 0.6 mm because the copper plates were much smaller. I didn't want it to get scraped off, so I opted for a smaller size. During drilling, I initially had issues using the dremel correctly and would constantly break the smaller drill bits, so I tried to go for a larger size as much as possible.



Figure F: Etching PCB in Ferric Chloride

This was taken in the middle of the etching process to show how the copper is eaten away. I did this to track my progress and ensure that everything around the mask was removed.

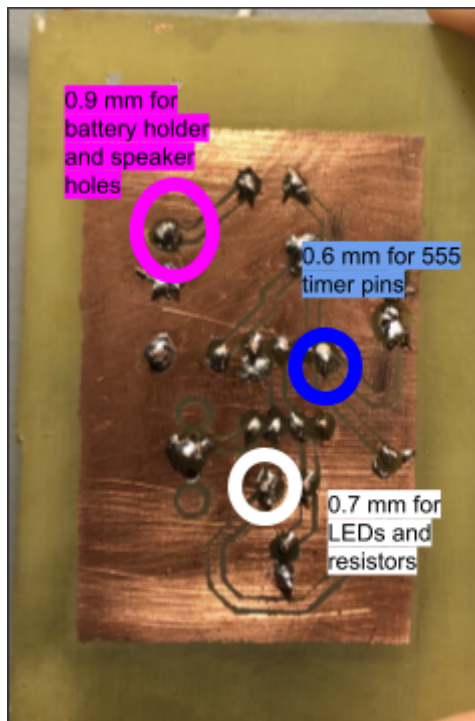
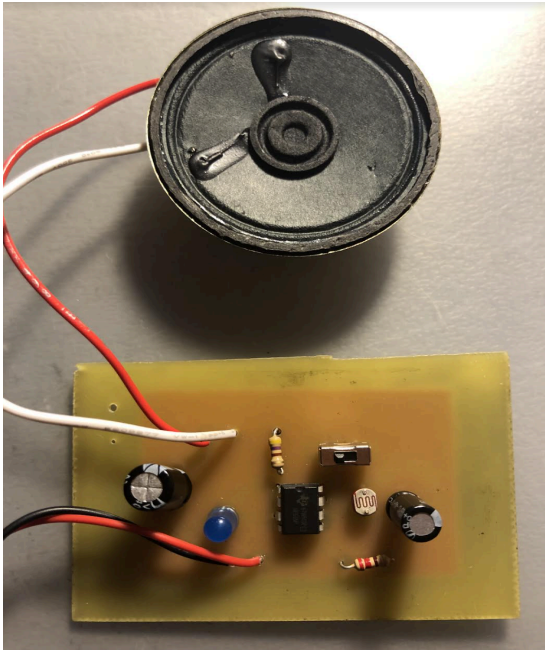
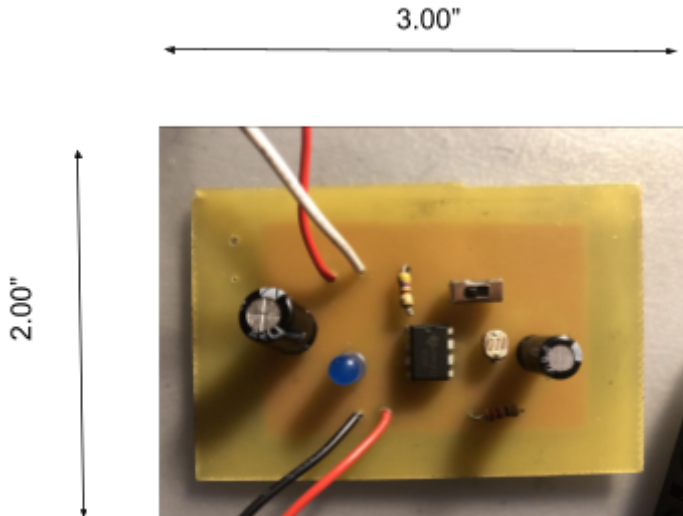


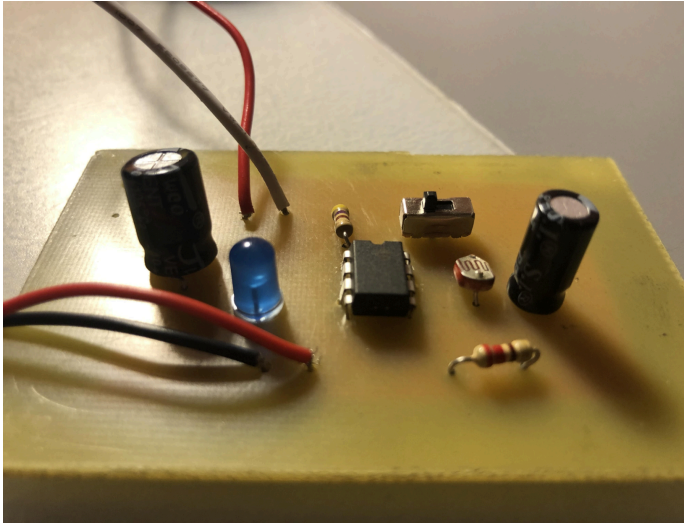
Figure G: Drilled and Etched PCB Board

This image shows the etched, drilled, and soldered PCB. It is difficult to see, but the holes covered in solder are 0.6 - 0.8 mm wide to fit the components and preserve the copper plates.

Solution

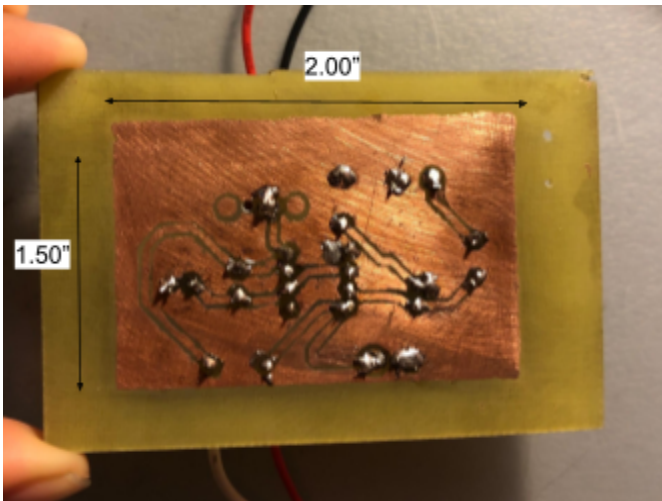
Final PCB

Image	Description
	Top View There is a speaker, battery connection, and soldered board. There are ten components in total. The board is unfortunately not functional, but the components are in their correct place, based on the PCB mask and schematic.
	Closeup Top View There is a 220 uF capacitor, 220 ohm resistor, 470 ohm resistor, 47 uF capacitor, LDR, 555 timer, slide switch, LED, 4.5 V power source, and speaker. The components match the design and fit right into their soldering holes. There was no bending or irregular placement of the components during the soldering step. Compared to the original copper board, this board is mostly yellow except for where the copper wires and ground plane are. The entire board is 2.00" x 3.00".



Front View

The front of the circuit shows the same components and their orientation on the board. They fit into the holes properly. However, the 555 timer was soldered directly into the board, as opposed to being placed on an IC socket, which may have caused issues with the functionality of the circuit.



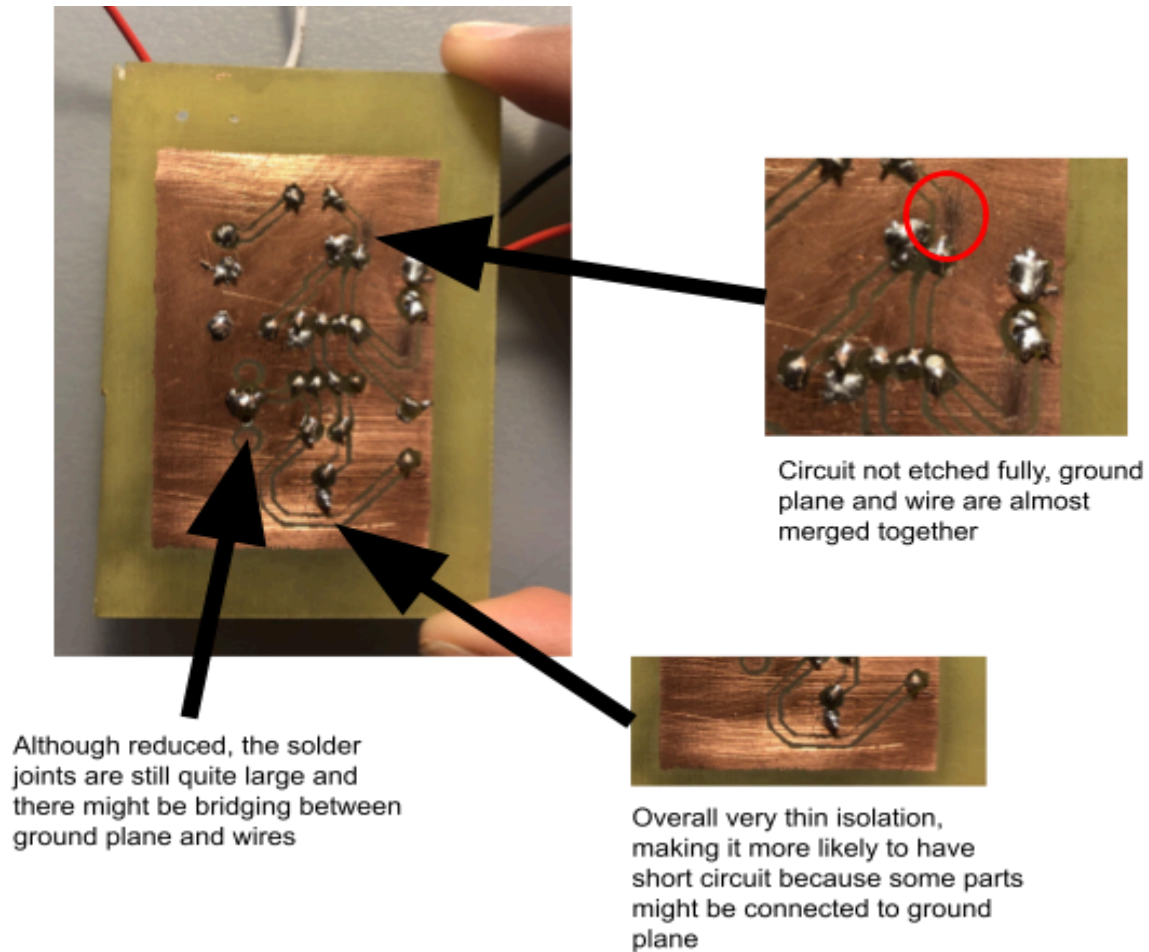
Bottom View (Solder Joints)

After drilling the holes, the components were soldered in. Due to the small size of many of my copper plates, my soldering was slightly extra in some areas, However, the joints are concave and there is a connection formed to the board. I had to reduce quite a lot of the solder sucker along the way because there were air bubbles that initially caused some to bridge.

The dimensions of the mask itself is 2.00" x 1.50".

Solution Description

Figure A: Analyzing Errors



My PCB board did not function; based on my observations, it most likely short circuited because the battery pack heated up very quickly when I connected it. This could be for several reasons, but the main two I found on my board could have been the lack of isolation for some of the wires and bridging between solder joints.

During etching, some of the isolations between my wires and the ground plane were not completed because the copper was not fully removed. The mask was also not dark enough in that area, so the ferric did not eat away at the copper. This meant that certain segments of the circuit that were powered would be connected to the ground plane. The ground plane would be powered as well, meaning that some components would be positively charged on both ends, creating a short/incomplete circuit. To tackle this, I attempted to use a sharp pin dipped in ferric chloride to scratch away at the small overlapping portions between the ground plane and wires. However, there still was a short circuit issue.

I checked that all my components were in the correct orientation, since swapping the positive and negative side of polarity sensitive components, like the LED and capacitor, could also result in a short circuit. After checking my schematic and PCB mask, it seemed my components were correctly in place.

Finally, I attempted to reduce the solder joints on my circuit, since bridging components to the ground plane may have resulted in a short circuit. I tested it after and the battery pack still got warmer, so I repeatedly removed solder and ensured that the components were only attached to their corresponding copper plate.

Due to time constraints, I was unable to fully determine what the cause of the short circuit was. I tried as many solutions as possible, but it appeared that I had not been careful during the etching and soldering processes, resulting in multiple issues. Failure to troubleshoot carefully along the way compromised the functionality of my circuit in the long run, which is an important lesson for future manufacturing projects. Although my circuit's functionality was not successful, I still learned a great deal about the PCB manufacturing process and the numerous steps involved. I picked up important skills in manufacturing and design, like using design software to create a circuit. Additionally, I learned that manufacturing is not a single step and it actually requires an incredible number of iterations to get an optimal design. I repeated masking, drilling, and soldering multiple times due to errors, and my circuit still did not function as intended. I also learned that in the earliest steps, being mindful of my design and manufacturing decisions will help make the rest of the steps easier. For example, if I had done a better job etching and drilling, my circuit would not have malfunctioned.

Citations

electronzapdotcom. (2021, September 6). *Light dependent resistor LDR astable 555 timer with speaker circuit learning electronics lesson 0064* [Video]. YouTube.

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Spurr, K. (2014, January 14). *How an LDR (Light Dependent Resistor) Works*. Kitronik Ltd.

<https://kitronik.co.uk/blogs/resources/how-an-ldr-light-dependent-resistor-works>